Massey University
College of Sciences

161.304: Comments and Solutions for Computer Laboratory 1

Exercise 1: R as a Calculator

1. $98765 - 43210 = 55555$
2. $12.3456789^2 = 152.4158$
3. $\sqrt{(1.23 + 4.56) \times (7.8 + 9.0)} = 9.862657$

1. `3+4*5` gives `23` (Note: R applies mathematical operations in the correct order.)
2. `3/4/5` gives `0.15`
3. `1:20/2` gives

```
 [1]  0.5  1.0  1.5  2.0  2.5  3.0  3.5  4.0  4.5  5.0  5.5  6.0  6.5  7.0  7.5
[16]  8.0  8.5  9.0  9.5 10.0
```

   Hence the sequence `1:20` is evaluated before the division is performed.

4. `(1:20)/2` gives the same answer as in the previous part.

Exercise 2: Exploratory Analysis of IBM Stock Price Data

1. No comment.
2. Mean and standard deviation computed by `mean(ibm)` and `sd(ibm)` respectively. Maximum and minimum can be derived from `summary(ibm)` or using `max()` and `min()` commands.
3. Element 249 of `ibm` had the decimal point omitted. This outlier can be located with `which(ibm>5000)`.
4. No comment.
5. `returns <- ibm[-1]/ibm[-250]`
6. `lreturn <- log(returns)`
7. The histogram suggests that the normal distribution is a very reasonable model for the (marginal) distribution of log returns.
8. `> t.test(lreturn,mu=0)`

```
        One Sample t-test

data:  lreturn
t = 0.0072, df = 248, p-value = 0.9942
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.002196363  0.002212568
sample estimates:
   mean of x
8.102561e-06
```

   No evidence that population mean log return is non-zero.

Exercise 3: Professor Kolmogorov Plays Cricket

1. The following code reads in the matrix data (by row, because I find that more natural, but there is no compulsion to do it that way).

```
> P <- matrix(c(0.4,0.2,0.2,0.1,0.1,0.3,0.3,0.2,0.1,0.1,0.2,
+ 0.2,0.3,0.2,0.1,0.1,0.2,0.3,0.2,0.2,0.1,0.1,0.2,0.4,0.2),
+ nrow=5,byrow=T)
>
> P
     [,1] [,2] [,3] [,4] [,5]
[1,]  0.4  0.2  0.2  0.1  0.1
[2,]  0.3  0.3  0.2  0.1  0.1
[3,]  0.2  0.2  0.3  0.2  0.1
[4,]  0.1  0.2  0.3  0.2  0.2
[5,]  0.1  0.1  0.2  0.4  0.2
```

2. The row sums are all 1:

```
> rowSums(P)
[1] 1 1 1 1 1
```

3. The initial distribution is $\boldsymbol{\pi}^{(0)} = (1, 0, 0, 0, 0)$ since the initial state is known to be $Y_0 = 1$. The code below computes $\mathsf{P}(Y_5 = 5|Y_0 = 1))$.

```
> pi0 <- t(c(1,0,0,0,0))
> library(expm)
> P5 <- P%^%5
> pi0%*%P5
        [,1]    [,2]    [,3]    [,4]    [,5]
[1,] 0.23788 0.20778 0.24212 0.18121 0.13101
```

   We conclude that there is a probability of 0.131 that his innings 5 will be excellent.

4. For $t = 10$ Professor Kolmogorov scores 20 runs, so $Y_{10} = 3$. It follows that $\boldsymbol{\pi}^{(10)} = (0, 0, 1, 0, 0)$. We now use the result that

$$\boldsymbol{\pi}^{(13)} = \boldsymbol{\pi}^{(10)} P^3.$$

   to compute probabilities for $t = 13$. Implementing this in R,

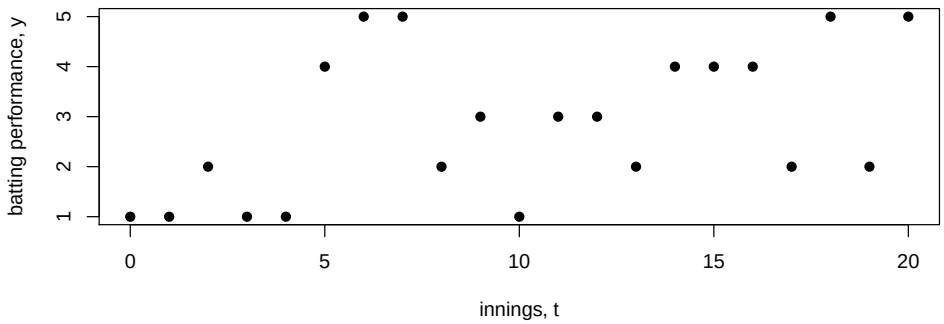
```
> pi10 <- t(c(0,0,1,0,0))
> P3 <- P%^%3
> pi10%*%P3
      [,1]  [,2]  [,3]  [,4]  [,5]
[1,] 0.236 0.208 0.243 0.182 0.131
```

   Hence the probability that he scores 30 runs or more, $\mathsf{P}(Y_{13} \geq 4) = 0.182 + 0.131 = 0.313$.

Exercise 4: Simulating Professor Kolmogorov's Cricket Season

The R matrix `P` is that produced in Exercise 3. Note that there is no need to sample Y_0 (which is stored as `y[1]`) because this is known to take value 1.

```
> y <- numeric(21)
> y[1] <- 1
>
> set.seed(87654321)
> for ( tt in 1:20 ){
+ y[tt+1] <- sample(1:5,size=1,prob=P[y[tt],])
+ }
> y
 [1] 1 1 2 1 1 4 5 5 2 3 1 3 3 2 4 4 4 2 5 2 5
> plot(0:20,y,pch=19,xlab="innings, t",ylab="batting performance, y")
```



We now move on to the multiple simulation. We first create an object to contain the results, in terms of the number of excellent (i.e. category 5) innings that Professor Kolmogorov plays. We then write a double loop to fill in the values. The inner loop simulates the Markov chain as above, while the outer one simply populates the results vector. Finally we computer the overall mean and its Monte Carlo standard error.

```
> results <- numeric(1000)
> set.seed(87654321)
>
> for (i in 1:1000){
+ y <- numeric(21)
+ y[1] <- 1
+       for ( tt in 1:20 ){
+       y[tt+1] <- sample(1:5,size=1,prob=P[y[tt],])
+       }
+ results[i] <- sum(y==5)
+ }
>
> mean(results)
[1] 2.609
> mc.se <- sd(results)/sqrt(1000)
> mc.se
[1] 0.05417614
```

The estimate mean number of excellent innings is 2.609 with a Monte Carlo standard error of 0.054.